

Gift Retail Sales & Customer Analysis

1. Project Overview

This project analyzes gift retail sales data to understand sales performance, customer spending, product performance, delivery efficiency, and order patterns.

The analysis was performed using Google Sheets, including data cleaning, formulas, Pivot Tables, and dashboard development.

The project focuses on identifying key business trends and generating actionable insights related to revenue, customers, products, occasions, and cities.

2. Dataset Summary

The project uses three datasets:

1. Customer Data

Rows: 100

Columns: 7

Key Features: Customer ID, Customer Name, City, Contact Number, Email, Gender, Address

The Customer dataset contains customer-level information used to analyze customer demographics, locations, and spending behavior.

2. Product Data

Rows: 70

Columns: 6

Key Features: Product ID, Product Name, Category, Price, Occasion, Description

The Product dataset contains product details used to analyze product performance, categories, pricing, and revenue contribution.

3. Orders Data

Rows: 1,000

Columns: 10

Key Features: Order ID, Customer ID, Product ID, Quantity, Order Date, Order Time, Delivery Date, Delivery Time, Location, Occasion

The Orders dataset contains transaction-level information and serves as the main dataset for sales and order analysis.

3. Exploratory Data Analysis (EDA)

Data Loading

Loaded the three datasets into Google Sheets and reviewed the structure, columns, and relationships between the datasets.

Initial Exploration

Reviewed the number of rows and columns, unique IDs, key fields, and relationships between Customers, Products, and Orders.

Verified that Customer ID and Product ID in the Orders dataset correctly matched the corresponding IDs in the Customer and Product datasets.

Missing Data Handling

Checked all relevant columns for missing values using blank-value checks.

No missing values were found in the three datasets.

Duplicate Data Handling

Checked Order IDs for duplicate records.

No duplicate Order IDs were found. Customer IDs and Product IDs were also verified to be unique.

Data Type Correction

Several columns were initially stored as text after importing the CSV files.

Corrected the data types for:

- Quantity → Numeric
- Price → Numeric
- Order Date → Date
- Order Time → Time
- Delivery Date → Date
- Delivery Time → Time

Customer IDs, Product IDs, Order IDs, and contact numbers were retained as text where appropriate.

Feature Engineering

Created additional features to make the analysis easier and more meaningful.

Revenue: Created by multiplying Order Quantity by Product Price to calculate revenue generated from each order.

Delivery Duration: Created using Order Date/Time and Delivery Date/Time to calculate the number of days taken to deliver each order.

Order Month: Created from Order Date in **YYYY-MM** format to analyze monthly sales trends.

4. Data Analysis & Insights

The analysis was performed based on the following business questions:

1. What is the total revenue generated?

Total revenue generated was **₹3,520,984** from 1,000 orders.

2. What is the average order value and average delivery time?

The average order value was **₹3,520.98**, while the average delivery duration was **5.50 days**.

3. How does monthly sales performance vary?

Sales performance varied significantly across months, with **February and August** recording particularly high revenue compared with several other months

4. Which products generate the highest revenue?

Product-level revenue analysis identified the highest-revenue products, with **Magnam Set** generating the highest revenue among the products analyzed.

5. Which customers contribute the most revenue?

Customer spending analysis was used to identify the **Top 10 customers by revenue**, helping highlight high-value customers.

6. What are the Top 5 products by revenue?

The Top 5 products by revenue were identified using product-level revenue aggregation and ranking.

7. Which cities have the highest number of orders?

The analysis identified the **Top 10 cities by order volume**, with **Dhanbad, Kavali, and Haridwar** among the highest-order cities.

8. Is there a relationship between order quantity and delivery duration?

The correlation between order quantity and delivery duration was **0.00415**, indicating that there was no meaningful linear relationship between the two variables in this dataset.

9. Which occasions generate the highest revenue?

Revenue by occasion analysis showed that **Anniversary, Raksha Bandhan, and All Occasions** generated the highest revenue among the listed occasions.

10. How does product popularity vary by occasion?

Product and occasion data were analyzed to understand product demand across different occasions and identify patterns in customer purchasing behavior.

5. Dashboard

Created an interactive **Gift Retail Sales & Customer Analysis** dashboard using Google Sheets/Excel-based analysis.

The dashboard includes:

Key Performance Indicators (KPIs)

- Total Orders: **1,000**
- Total Revenue: **₹3,520,984**
- Average Order Value: **₹3,520.98**
- Average Delivery Time: **5.50 days**

Dashboard Visualizations

- Revenue by Category
- Top 5 Products by Revenue
- Top 10 Customers by Revenue
- Revenue by Month
- Revenue by Occasion
- Top 10 Cities by Orders

Filters

- Occasion
- Category

The dashboard provides a consolidated view of sales, customer, product, and order performance to support business-level analysis.



6. Business Recommendations

1. Prioritize high-performing product categories

Colors generated ₹1,005,645, the highest category revenue and approximately **28.6% of total revenue**. Soft Toys and Sweets followed with ₹740,831 and ₹733,842. These categories can receive greater promotional and inventory focus.

2. Plan campaigns around high-revenue occasions

Anniversary generated ₹674,634, followed by Raksha Bandhan at ₹631,585 and Holi at ₹574,682. Seasonal campaigns, product bundles, and inventory planning can be strengthened around these occasions.

3. Use high-performing months for campaign planning

August generated the highest monthly revenue at ₹737,389, followed by February at ₹704,509 and March at ₹511,823. These periods can be studied further to understand the products and occasions driving higher sales and to plan inventory accordingly.

4. Target high-value customers

The Top 10 Customers by Revenue analysis identifies customers contributing the most revenue. These customers can be targeted with personalized offers, loyalty benefits, and repeat-purchase campaigns.

5. Focus on high-order-volume cities

Dhanbad, Kavali, and Haridwar were among the cities with the highest number of orders. These markets can be monitored for demand patterns and targeted promotional activity.

6. Investigate delivery efficiency separately from order quantity

The correlation between Quantity and Delivery Duration was only **0.00415**, indicating no meaningful linear relationship in this dataset. Therefore, delivery duration should

be investigated using other factors such as location, product, or occasion rather than assuming larger orders cause longer delivery times.